Streaming speech translation on a Raspberry Pi 5

Fredrik Boglind Tiny Machine Learning, autumn 2026

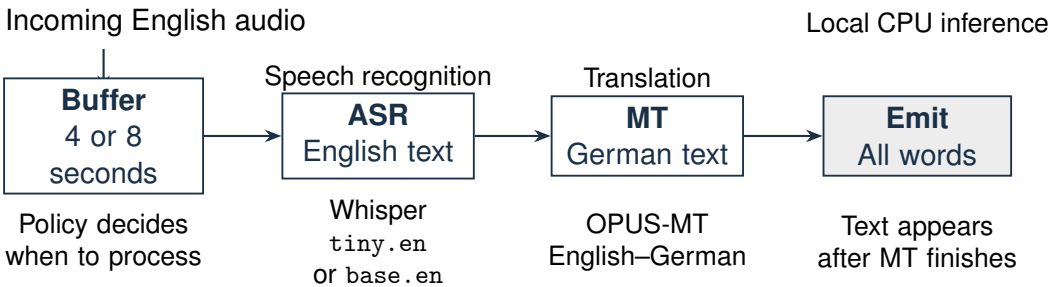
Research questions

1. How does computation change the quality–latency trade-off?
2. Which configurations keep up with incoming speech, and how does the Pi compare with a laptop?

English speech to German text. Submitted report, Chapter 1.

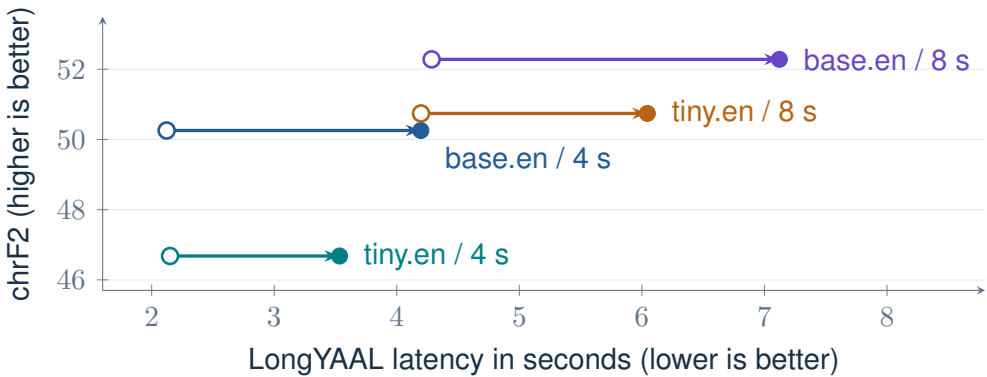
What I built and evaluated: A cascaded SimulST system

Simultaneous speech translation produces text while speech is still arriving.



Evaluation: Raspberry Pi 5 and ThinkPad CPU; 6 MCIF talks (33.6 minutes).
96 runs: 2 ASR models $\times$ 2 chunk sizes $\times$ 2 devices $\times$ 6 talks $\times$ 2 repetitions.
Additional work: Whistle comparison, error review and a LocalAgreement prototype

Computation increases measured latency

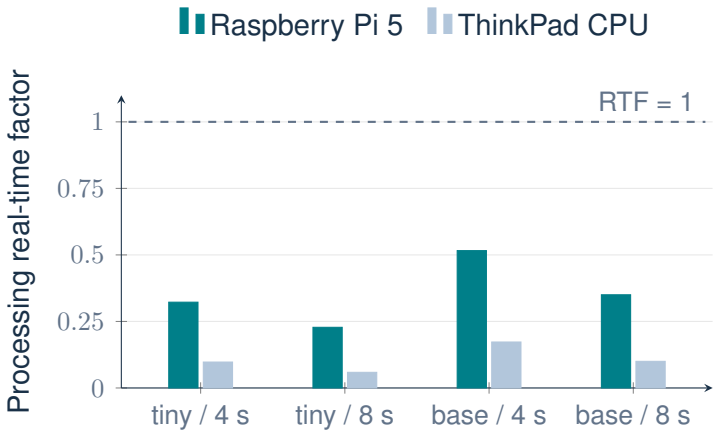


Hollow: computation ignored **Filled:** computation included

Longer chunks and the larger ASR model improve quality, with more delay.

Raspberry Pi results across six MCIF talks. Submitted report, Table 4.1.

All Pi configurations kept up on average



Pi RTF: 0.23–0.52

The Pi needed about 3–4× the laptop's processing time.

8.85 s

Largest Pi queue wait. Repetitive output temporarily blocked later chunks.

Submitted report, Tables 4.1–4.2. RTF = processing time / audio duration. Loading and warm-up was excluded.

4-second vs 8-second chunking

MCIF · RECORDED PI RUNS

Raspberry Pi 5 · Whisper tiny.en → Marian English–German · 1× playback

Audio 00:56.0



4-second chunks

Recognise and translate each chunk once

Audio ASR MT — Collecting audio

Current chunk: 52–56 s — 4.0 / 4 s received

LATEST ENGLISH RECOGNITION

which is the notion that multifaceted social identities can...

Available at 00:52.87

GERMAN OUTPUT — Latest: 00:53.20

Vereine wie negative Assoziationen mit bestimmten Gruppen. Darüber hinaus sind die meisten Arbeiten in diesem Raum nicht für Kreuzung. **Das ist die Vorstellung, dass vielfältige soziale Identitäten...**

8-second chunks

Recognise and translate each chunk once

Audio ASR MT — Collecting audio

Current chunk: 48–56 s — 8.0 / 8 s received

LATEST ENGLISH RECOGNITION

associations like negative associations with particular groups. Furthermore, most work in this space doesn't account for intersection

Available at 00:49.03

GERMAN OUTPUT — Latest: 00:49.72

... einfach eine sehr allgemeine Rute ein. **Assoziationen wie negative Assoziationen mit bestimmten Gruppen. Darüber hinaus sind die meisten Arbeiten in diesem Raum nicht für Kreuzung**

Same audio and models · Only chunk length changes · Separate measured runs

Latest German output in bold

Supplementary work

Whistle comparison

Lower sampled Pi memory, but higher total RTF than tiny.en.
RTF about 0.40–0.42; much of the time was spent in MT.

Exploratory error review

Jev screening, a second model review and limited manual inspection found cases worth investigating.

Threshold sensitivity and evaluator disagreement prevent treating the counts as error rates.

Submitted report, Sections 4.4–4.5 and 5.3. Whistle used a different inference implementation; this is a system comparison.

Problems and limits

- Chunk boundaries and recognition repetition can damage meaning and delay output.
- Six prerecorded scientific talks in one language pair limit generalisation.
- The quality review was exploratory, without expert bilingual validation.
- Live conversational/practical use was not evaluated.

Submitted report, Sections 4.3–4.4 and 5.4. Repeated runs do not add independent test material.

Answers and main takeaway

How does computation change the trade-off?

Pi latency rises to **3.53–7.12 s** when computation is included.
More context improves quality, but increases delay.

Can the system keep up?

All four main Pi configurations can, on average.
The laptop is faster, and temporary stalls remain possible.

A reproducible SimulST workload for comparing deployment environments (not systems)

Computation-aware evaluation shows how the same translation configuration behaves on the Pi and laptop, considering quality, delay and processing capacity together. **A benchmark?**

Submitted report, Chapters 5–6. Latency values are computation-aware LongYAAL, not first-output times.

Backup: Chunk length: waiting time and context

4-second chunks

More frequent output, less context

English recognition

“**over at Lee Negative** or Toxic. . . ”

German translation

“**bei Lee Negative** oder Giftige. . . ”

8-second chunks

Longer wait for audio, more context

English recognition

“. . . outputs **aren’t overtly negative** or toxic. . . ”

German translation

“. . . Ausgänge sind **nicht offen negativ** oder giftig. . . ”

The recognition error carries into German.
The 8-second chunk retains the negation in this example.

Verbatim excerpts from the matched Pi tiny.en runs, first repetition. Video time 00:42.

Backup: all main configurations

Device	ASR / chunk	chrF2	COMET	WER %	CU (s)	CA (s)	RTF
Pi	tiny.en / 4 s	46.68	0.559	36.02	2.15	3.53	0.322
Pi	tiny.en / 8 s	50.74	0.626	25.61	4.20	6.04	0.228
Pi	base.en / 4 s	50.26	0.602	26.36	2.12	4.20	0.516
Pi	base.en / 8 s	52.28	0.659	22.92	4.28	7.12	0.350
Laptop	tiny.en / 4 s	45.53	0.556	40.91	2.19	2.59	0.097
Laptop	tiny.en / 8 s	50.52	0.629	26.18	4.20	4.67	0.059
Laptop	base.en / 4 s	49.92	0.603	26.95	2.13	2.82	0.173
Laptop	base.en / 8 s	51.31	0.650	29.39	4.22	5.02	0.100

Quality: corpus chrF2, segment-mean COMET, pooled WER. Timing: mean of two repetitions. CU and CA use different output clocks.

Submitted report, Table 4.1. Six talks, 33.63 minutes, 266 reference segments.